6. MASS AND BALANCE

6.1 EQUIPMENT LIST ENTRIES

System/Equipment Designation	Part No.	Mass (kg)	Arm (mm)	Massmoment (kgmm)
Automatic Flight Control System, Sperry HELIPILOT	DSK3-50573			
– Computer, HELIPILOT	4025008-914	2.7	4300	11610
Automatic Flight Control System, Sperry HELIPILOT	D134-4277			
– Computer, HELIPILOT	4015985-904	4.4	4300	18920

7. SYSTEM DESCRIPTION

The HELIPILOT Automatic Flight Control System (AFCS) is a 3-axis control system and can be operated by either “hands-on” or “hands-off”.

The HELIPILOT System and optional HELCIS Flight Director System are not coupled so that the HELIPILOT System operation is restricted in its function as a stability augmentation system in SAS or attitude hold modes.

The HELCIS Flight Director is only used to provide roll, pitch and collective commands which are displayed on the pilot’s ADI. (For further information on the HELCIS system refer to the AVM).

The AFCS achieves stability augmentation by sensing pitch/roll attitude and yaw rate of the helicopter and generating control signals for the connected actuators. The control authority of the system with both helipilot computers functioning is limited as follows:

Pitch axis: 2 actuators (2 x ± 5 %) = ± 10 %
Roll axis: 2 actuators (2 x ± 10 %) = ± 20 %
Yaw axis: 1 actuator (1 x ± 10 %) = ± 10 %

7.1 AUTOPILOT MODES

7.1.1 Attitude Hold (ATT)

The function ATT is an attitude retention mode, which maintains the attitude of the helicopter.

The attitude hold mode is the basic mode of operation and is selected automatically when engaging either of the helipilot computers (HP 1 or HP 2) because the solenoid operated ATT/SAS switch will spring to the ATT position when both computers are disengaged.

The attitude hold mode provides the following functions:

– Damps external disturbances
– Provides attitude retention with attitude response to cyclic.

7.1.2 Stability Augmentation (SAS)

The function SAS is a stabilization mode designed to ease the pilot workload during normal flight.

The SAS mode is selected by placing the ATT/SAS switch to the SAS position. There is no further indication that shows that SAS is selected. The SAS mode will be cancelled when selecting the ATT mode.

The SAS mode provides the following functions:

– Damps external disturbances (i.e. gusts)
– Reduces control cross-coupling
– Provides attitude rate-response to cyclic
– Provides short term attitude stability.

7.1.3 YAW SAS (YAW) Mode

Following functions are provided by the YAW mode:

– Critically damped Dutch Roll under all flight conditions
– Feet-on-floor trim capability under all flight conditions.

7.2 SYSTEM TRIMMING

The actuator position indicators (API) give an indication of series actuator position.

During flight phases when the helicopter is being controlled by the AFCS computers, there are times when it will be necessary for the pilot to momentarily be in the control loop, e.g. in turbulence, during power changes or when large flight maneuvers have been commanded. The pilot is alerted to the need for manual trimming when one or more of the APIs indicate a steady deflection from center.

This condition can easily be corrected by trimming the control system with the four-way trim switch located on the top of the cyclic stick grip or with the yaw trim switch located on the pilot's collective lever.

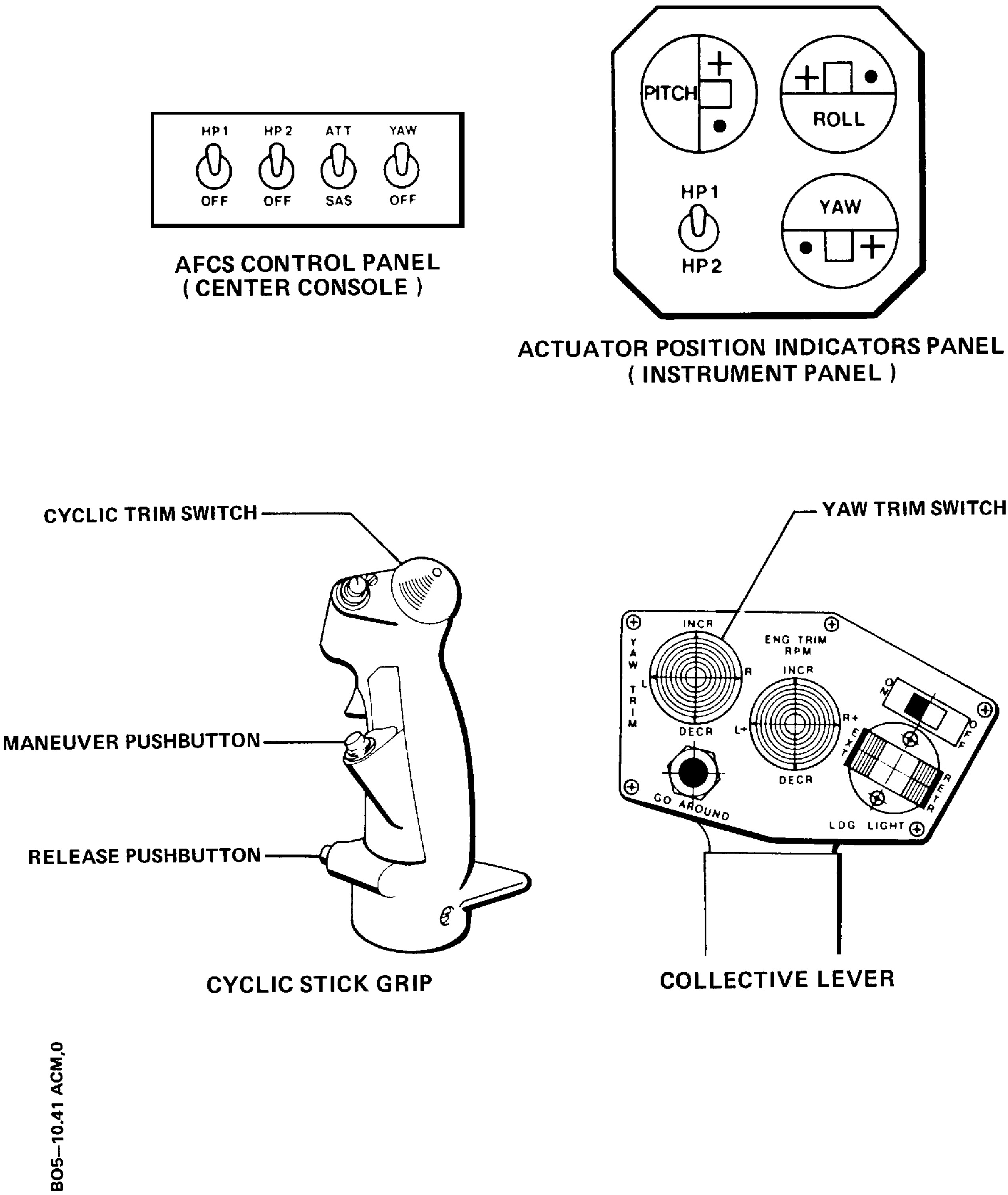

Fig. 2 AFCS Controls and Indicators